

urothelial-mets-her2-discordant-kndl

Plain-language summary for patient and caregiver

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What this page is

Libby searched published studies and open clinical trials for treatments aimed at the specific features of your cancer, the ones in your records. This page walks through what came back: one first step that everything else depends on, ten treatment options worth discussing with your care team, four options you may hear about that are on the page as not recommended, and a longer tail of possibilities that were looked at and set aside, with the reasons why. Where the research has real numbers, we give them in plain terms. Where it does not, we say so instead of dressing anything up.

One honest note before anything else. Almost every option on this page hinges on test results that already exist but have not been pulled out of their reports. That is unusual, and it is actually good news: the most important next move in your case costs a records request, not a treatment decision.

What we know about your cancer

You have urothelial (bladder) cancer, high grade, that had spread to the liver when it was found in late 2023. First-line treatment with an antibody-drug combination plus immunotherapy worked remarkably well: the liver disease resolved completely, and that response held for about two years. The antibody-drug part was stopped after roughly 25 months because of nerve damage in your hands and feet, and this year the cancer came back in two lymph nodes in your abdomen while you were on maintenance immunotherapy. Those nodes were treated with focused radiation in June. You started a new drug, trastuzumab deruxtecan (T-DXd), in July; it targets a protein called HER2 and you have had two doses, with severe nausea and bloating after the second. You also have a separate, earlier prostate cancer that was treated in 2020 and shows no sign of activity.

The single most important fact on this page is a disagreement between test results. Three separate high-quality tests, on tumor tissue from 2023 through spring of last year, all found HER2 at the strongest level (called 3+). But a research-grade imaging test run this month on the newest biopsy found only a weak HER2 signal. The research test is not an approved way to measure HER2, so it cannot overrule the three strong results, but it raises a real question: does the cancer as it exists today still carry high HER2? A proper repeat HER2 test was already done on that new tissue this month. The answer is sitting in a report that has not been retrieved yet.

Two more things sit unread in existing reports. Your 2023 tumor sequencing counted exactly one genetic change with a matching treatment, and the report in hand never says which one it is. And your records never state whether the tumor carries a change in a gene called FGFR3, which about 15 to 20 percent of cancers like yours have and which has its own approved pill.

Other features of your tumor matter for the options below. It carries very high levels of a surface protein called Nectin-4, at least it did before treatment; the drug that produced your two-year response targets exactly that protein, and long exposure can teach a tumor to dial it down, so today's level is unknown. A research test found low levels of a protein called TROP2 and strong signals for two others (mesothelin and MUC1), all needing confirmation with standard tests. The tumor has a higher-than-average number of mutations, which can favor immunotherapy. Blood tests found a trace of a mutation called TP53 R175H, and you carry the immune-system type (HLA-A*02:01) that some newer trials require. A blood test that tracks tumor DNA has been positive for a long time, rose this spring, and then read zero on two draws after the radiation, which may mean the radiation did real work; the next draws will tell.

Two health flags shape the choices below: one chest scan last year suggested possible early lung scarring while a

more recent scan showed none (an unresolved discrepancy that matters because some drugs, including your current one, can inflame the lungs), and you had a mild, symptom-free bout of pancreas inflammation in 2025 that your team still monitors with blood tests.

What you told us matters most

Here we have to be straightforward: you did not fill out a preferences form. Everything in this section was inferred from your records, so treat it as a set of defaults to confirm or correct, not as your words.

The record reads as strongly experimental-leaning. You are already pursuing personalized cancer vaccines, cell-therapy programs, and other investigational routes at the same time, and you have traveled internationally for testing and consults. So this page takes trials seriously as first choices, not fallbacks, and it weights effectiveness somewhat more heavily than side effects.

Four side-effect limits were inferred, and they do real work in the ranking below. First, drugs that cause nerve damage: you already have lasting numbness and hand spasms from prior treatment, so options built on that drug class were pushed down or off the list. Second, drugs that cause severe nausea and gut trouble, because that is an active problem on your current drug. Third, drugs with high lung-inflammation risk, because of the unresolved chest-scan question. Fourth, drugs that stress the pancreas, because of the 2025 episode. One practical constraint also matters throughout: your remaining tumor tissue is scarce, and the newest biopsy is only 5 to 10 percent tumor cells, so options that can be screened with blood tests, scans, or existing records were favored over ones that demand a fresh biopsy.

The first step everyone agreed on

Before any treatment decision, two sets of existing results need to be retrieved and read. This was the one step with unanimous support, it involves no drug and no procedure, and everything below it hangs on the answers.

The first is the HER2 answer. The repeat, properly validated HER2 test on your newest tissue was already performed this month; only the result tier has not been pulled into your record. Retrieving it is a records request that should take days. If the tissue has to be recut and restained instead, that is about three to seven days plus paperwork. One caution to carry into the reading: the newest biopsy contains only 5 to 10 percent tumor cells, which can make a surface-protein stain read weaker than it really is. Ask that the report state the tumor content, and a second read on the tissue from last spring's recurrence is a sensible hedge against that sampling problem.

The second is the sequencing retrieval. Your existing tumor-sequencing reports, from 2023 through 2025, hold three unread answers: whether the tumor has an FGFR3 gene change (roughly a 15 to 20 percent chance, and if yes it opens a treatment door with formal guideline backing, Option 6 below), the identity of the one genetic change your 2023 report says has a matching treatment, and the tumor's HER2 gene copy number, which one of the HER2 options below turns on. All of this comes from reports that already exist. No new tissue is spent.

Why this gates everything: the HER2 result decides three of the options below, and it feeds the separate decision about whether to continue your current drug, whose approval for your situation is written for the strongest HER2 level. Study after study shows the benefit of HER2 drugs tracks the HER2 level closely; in the key trial behind your current drug, about 6 in 10 people at the strongest level saw their tumors shrink, versus fewer than 4 in 10 across all levels. And the sequencing answer could surface an option that outranks most of this page.

What if the HER2 test comes back negative or low? Then the three HER2 options below (Options 1, 5, and 7) come off the table, and your current drug loses its formal approval footing for you; it would then rest on weaker evidence borrowed from other cancer types, which is a different and less certain conversation. The rest of the options on this page do not depend on HER2 and stand or fall on their own tests. If those other doors were also to close, no retained Nectin-4, no confirmed TP53 mutation in tissue, no FGFR3 change, and negative confirmatory stains on the research leads, then this page would have nothing further to rank, because Libby's list only covers treatments aimed at your cancer's specific targetable features. Standard treatment for bladder cancer would still exist and would still be worth pursuing, but that is a separate conversation that belongs with your oncologist, not on this page.

The options

Every option here is investigational for your situation: a trial drug, an off-label use, or an early-stage program. The decision about continuing your current drug, T-DXd, is deliberately not ranked here, because it is an approved treatment and belongs to the standard-care conversation with your team. What this list answers is: if T-DXd stops, or when you and your team look beyond it, what targets your cancer's specific features?

Option 1 — Zanidatamab (a HER2 antibody with no chemotherapy payload)

This option is only on the table if the HER2 test comes back positive at the strongest level. If it comes back negative or low, this option is off the table, along with Options 5 and 7. The rest of the page does not depend on that test.

This option carries caveats. Zanidatamab is an antibody that grabs the HER2 protein at two points at once. Unlike your current drug, it carries no chemotherapy payload, and that is the whole point of its position here: in its first human study, severe drug-related side effects occurred in only 4 of 132 people and nobody died from treatment. It touches none of the four side-effect limits above. If the strong HER2 result holds but the current drug has to stop because of the nausea, this is the gentlest way to keep treating the target.

The tradeoffs are real. In that same study, about 1 in 3 people saw their tumors shrink, but most of those people had bile-duct cancer, not bladder cancer, and nobody has measured how the drug performs after a drug like your current one. There is also a biological worry: if the cancer survived a HER2-targeted drug by reducing its HER2, an antibody with no payload could be hit even harder by that change. And the practical route is awkward, because the trial designed for this situation excludes people who have already had HER2-directed treatment, which your current drug counts as. The realistic path is off-label prescription, and insurance approval would lean heavily on that HER2 result.

Option 2 — AKY-1189 (a radioactive particle aimed at Nectin-4)

This option carries caveats: it is a first-in-human study with no results of any kind yet. This treatment attaches a powerful, short-range radioactive particle to a molecule that seeks out Nectin-4, the protein your most effective treatment so far was aimed at. The radiation kills the cell it lands on and its close neighbors, so it matters less if some tumor cells have reduced the protein. That is the appeal in your exact situation: after 25 months of a Nectin-4 drug, some loss of the target is plausible, and this mechanism is the one that best survives it. The study also starts with an imaging scan that shows whether your tumor still carries Nectin-4 today, answering that open question with a scan instead of spending scarce tissue. Prior treatment with your previous drug is explicitly allowed, which almost no other Nectin-4 study permits.

Be clear about what is not known: everything. No person has published results on this drug, for benefit or for harm. The known concerns of this drug class are bone-marrow suppression (you had anemia in 2025, since corrected) and stress on the kidneys, and your current kidney-function number is missing from the record, a gap that needs closing before this conversation goes anywhere. Early joiners in a dose-finding study may also receive doses too low to work, because finding the dose is the study's purpose. Two screening hurdles apply: the study needs a measurable tumor on scans, and the radiation you had in June treated the only known sites. Sites are within reach in the Northeast. A sensible first move that costs nothing: retrieve the result of the Nectin-4 imaging you already had in June, which may answer the expression question before any screening visit.

Option 3 — CLSP-1025 (a drug that points your immune cells at a TP53 mutation)

This option carries caveats: the drug class has no human results yet, and the mutation it targets has only been seen in your blood, at a very low level. This is an off-the-shelf drug (no cell collection, no manufacturing wait) that acts like a matchmaker: one end recognizes a fragment of the mutated TP53 protein displayed on tumor cells, the other end grabs your immune cells and forces the two together. It requires two things you already have on paper: the specific mutation (TP53 R175H) and the immune-system type HLA-A*02:01. A mutation like this sits deep in the cancer's machinery, so the tumor cannot easily shed it the way it can shed a surface protein. The one human precedent for attacking this exact target, with a different technology, was a single patient whose tumors shrank by half for six months.

Now the hedges, and they are load-bearing. No drug of this class has published results in people, so benefit and safety are both unmeasured; the expected side effect is an immune overreaction called cytokine release syndrome, with the first doses given in the hospital. And the mutation call itself is the weak link: it comes from a single blood test that found it in 0.09 percent of the DNA fragments examined. A trace that small is a much less sure sign than finding the mutation in the tumor itself; it could be a minor subpopulation, or it might not confirm at all when the trial's lab checks. Before any screening visit, ask your team to look for this mutation in your existing 2023 sequencing report. That is free, fast, and could firm up or quietly close this option.

Option 4 — LY4052031 (a next-generation Nectin-4 drug, built for people in your exact situation)

This option carries caveats. This is an antibody-drug conjugate like the one that gave you your two-year response, aimed at the same Nectin-4 protein, but carrying a different chemotherapy payload, one that does not cause the nerve damage that forced your previous drug to stop. The study group in question was written specifically for people who already had that previous drug; prior exposure is required, not excluded. The biology also leans favorable: studies of how tumors escape your previous drug suggest they more often break the delivery machinery than discard the Nectin-4 target itself.

The headline number needs honest handling. Early results reported at a conference this year showed tumors shrinking in about half of previously treated patients at the chosen dose. That figure is unpublished, comes from a small ongoing dose-finding study run by the company, and has no error bars; numbers like this usually shrink as more patients are added. Treat it as a reason to screen, not a promise. There is also no published side-effect table yet; the payload class typically causes diarrhea, nausea, and low blood counts, which lands on your active nausea problem and needs a plan rather than a shrug. Whether a payload of this type works well right after your current drug, whose payload is a chemical cousin, is an unstudied question worth putting to the trial doctor directly. Several sites are within reach in the Northeast, screening runs on records rather than fresh tissue, and the same two hurdles as Option 2 apply: measurable disease after the June radiation, and a washout period after your

current drug.

Option 5 — Pertuzumab plus trastuzumab (two well-known HER2 antibodies, no payload)

This option is only on the table if the HER2 test comes back positive, at the strongest level or with extra HER2 gene copies on the retrieved report. If the test reads low without extra copies, this option closes.

This option carries caveats. These are the two oldest, best-understood HER2 antibodies, used together without any chemotherapy attached, available through a structured access study or off-label. The supporting study is the clearest illustration of why the HER2 test matters so much: tumors shrank in about 4 in 10 people at the strongest HER2 level with extra gene copies, about 1 in 4 overall, fewer than 1 in 10 at the weak level, and nobody at zero. Two of the five complete responses in that study were bladder cancers. The side-effect profile is the mildest of any drug option on this page: diarrhea in about a third of people, mostly mild, with more than a decade of safety experience behind both drugs and routine heart monitoring the only standing requirement. It is the one HER2 option that touches none of your four side-effect limits.

The open questions: that study was not randomized, benefit was measured by tumor shrinkage rather than survival, and nobody has tested these antibodies after a drug like your current one, so a tumor that dodged your current drug by reducing HER2 would likely dodge this too. What is contested about its placement is a values question worth settling yourself: it trades the higher, shakier upside of the newer trial options for a modest, better-documented benefit with gentler side effects. Where it sits in your personal order depends on how you weigh that trade, and on two results from the first step: the HER2 tier and the gene copy number.

Option 6 — The FGFR3 route: a records request away from real

This option carries caveats: it only exists if the sequencing retrieval finds an FGFR3 gene change, which is roughly a 15 to 20 percent chance. If the change is there, this becomes one of the strongest doors in the case, and the only non-HER2 route with formal guideline backing for exactly your situation. There is an approved daily pill for FGFR-altered bladder cancer (that belongs to the standard-care conversation with your team), and there is a trial of a next-generation FGFR3-selective drug for the investigational route; one of its study groups is written for people who have both an FGFR3 change and HER2 expression, the single spot in the trial registry where your two main features meet.

Nothing about your tumor is known to argue for or against the change being present; the answer is simply unread. The typical side effects of this drug class are high blood phosphate, mouth sores, and nail and eye problems, none of which touch your four limits. The honest arithmetic: an 80 to 85 percent chance the retrieval closes this door, a 15 to 20 percent chance it opens something with more consensus behind it than most of this page. The records request that answers it is the same one described in the first step.

Option 7 — FT825 (an off-the-shelf HER2 cell therapy)

This option is only on the table if the HER2 test comes back positive at the trial's required level. If not, it closes along with Options 1 and 5.

This option carries caveats: nothing has been measured for it in bladder cancer. This is a cell therapy grown from donor-derived stem cells and engineered to hunt HER2, given after a short course of chemotherapy that clears space for the cells. Because it is off-the-shelf, there is no cell collection and no manufacturing wait, and unusually, prior HER2 treatment does not exclude you, which makes it one of very few HER2 trial doors your

current drug has not already shut. It avoids the payloads behind your nausea and the lung-inflammation worry.

The costs are structural. The clearing chemotherapy brings a window of low blood counts, the therapy can trigger the same immune overreaction described under Option 3, and this is an early dose-finding study with no results yet in your cancer type. One quieter concern deserves a direct question: the clearing chemotherapy suppresses immune cells broadly, and you have spent a year building up vaccine-trained immune cells; whether one would undo the other should be asked before anything is scheduled. Also ask whether the ongoing vaccine series trips the trial's exclusion for prior cell or gene therapy. Sites are within your usual care orbit.

Option 8 — ASP2998 (a TROP2-aimed drug that carries an immune alarm instead of chemotherapy)

This option carries caveats, and the biggest one is your own test result. This early trial drug delivers an immune-alarm signal (a "STING agonist") to cells carrying the TROP2 protein, waking the immune system inside the tumor rather than poisoning cells directly. Bladder cancer is a named study group, there is no cap on prior treatments, and prior TROP2- or similar-payload therapy is explicitly allowed, so your current drug does not close this door the way it closes most TROP2 doors.

The hedge that has to travel with this option: the research-grade test this month found your TROP2 level to be low. A low or weak result is a real positive, but it is a much less sure sign that a TROP2-aimed drug will help than a strong result would be, and that makes this option more uncertain than its neighbors. A validated TROP2 test was already run this month; its tier is in the same unretrieved stack as the HER2 result, and it should be read before this option gets real weight. Beyond that, the drug has no published results on benefit or safety. An immune-stimulating payload in someone with a 2025 bout of pancreas inflammation means the existing blood-test monitoring belongs in the plan. And one arm of this trial combines the drug with the agent that caused your nerve damage; the acceptable arms (drug alone, or with pembrolizumab) should be settled with the site in writing before consent. Sites are within reach.

Option 9 — Adding ipilimumab to restart immunotherapy, at a reduced dose only

This option carries caveats, and it is the most contested item on this page. The idea: your tumor has an above-average mutation count and once responded deeply to immunotherapy, so adding a second immune drug (ipilimumab, which releases a different brake than the one your previous immunotherapy released) might reignite that response. Your care team is already exploring this, and it is the only option here that could start now, with no screening test, no trial slot, and no washout debate.

Here is the tension, plainly. In the trial that tested this exact maneuver after progression on a drug like yours, about 1 in 3 people responded, but that result came with serious costs: about 1 in 9 developed severe immune inflammation of the gut, and 2 of 83 died from treatment. That full-dose schedule collides with two of your four side-effect limits at once (severe gut toxicity, and pancreas risk given your 2025 inflammation), and it is off the table on this page. What is ranked instead is a reduced-ipilimumab schedule with mandatory pancreas blood-test monitoring and written stopping rules. The honest catch, and the reason this option is contested rather than settled: every good response number in this drug class came from the full dose. The gentler schedule has never beaten immunotherapy alone in any trial, so you would be accepting added immune-side-effect risk for a benefit that is plausible but unproven. It also has no formal guideline listing for bladder cancer, which makes both the safety scaffolding and the insurance case things your team would have to build themselves. A registered trial exists that includes this combination, but its entry rules require a tumor type composition yours probably does not

meet. If this option appeals, the questions section below has the specific things to pin down first.

Option 10 — Sacituzumab tirumotecan trial (a randomized TROP2 study)

This option was considered but not recommended. It's on the page because Libby surfaces what was weighed and rejected, not just what was kept. This is a large randomized trial of a new TROP2 drug, and randomized entry is normally the cleanest, most rigorous way into a new drug class. It fails here for two independent reasons. First, the comparison arm is chemotherapy, and two of the three allowed chemotherapy drugs are from the class that caused your lasting nerve damage; joining means accepting a coin flip that could assign you a drug your situation forbids, with no opt-out. Second, you are likely not eligible anyway: the trial requires prior platinum chemotherapy, a treatment you never had. That platinum requirement recurs across several trials in this space and is worth flagging to your team as a strategic question in its own right. The class history adds a further reason for caution, described under Option 13.

Option 11 — Continuing your personalized vaccine programs

This option carries caveats, and the honest label for it is: hypothesis stage. You are already receiving personalized vaccines built from your own tumor's mutations, across two programs, with blood-based monitoring that fits the tissue-scarcity constraint. This entry is about continuing, not starting, and about being straight regarding what the evidence shows so far. In your own monitoring, immune responses appeared against 2 of 14 evaluable vaccine targets at four months, of a helper type rather than the killer type that most directly attacks tumors, and your tumor-DNA blood test rose through most of the vaccine period. No tumor shrinkage in your record is attributable to vaccination. The encouraging randomized evidence for this vaccine class comes from a different cancer (melanoma) in a different setting.

Why it stays on the page: side effects are the mildest here (sore arms, feeling run down), it is fully aligned with the path you have chosen, and the banked immune cells it generates feed the future cell-therapy programs you are pursuing. The immune-response test after the current mRNA series finishes is a named branch point: it will tell you whether this axis is building something or not. One interaction to manage deliberately: ongoing vaccine dosing may count as "prior gene therapy" under some cell-therapy trial rules, including Option 7's, so ask before it quietly closes a door.

Option 12 — Trastuzumab plus paclitaxel

This option was considered but not recommended. This off-label HER2 combination pairs an antibody with a taxane chemotherapy. In the supporting study, nearly 9 in 10 people developed some degree of nerve damage from the taxane. You already have lasting nerve damage from prior treatment, and gentler ways to target HER2 exist on this page (Options 1 and 5). The rejection stands regardless of how the HER2 test comes back.

Option 13 — Sacituzumab govitecan

This option was considered but not recommended. This TROP2 drug once had a conditional approval in bladder cancer. In its earlier study, tumors shrank in about 1 in 4 people, but the larger confirmatory trial found that people did not live meaningfully longer than with standard chemotherapy, about 7 in 100 died of treatment-related complications (mostly infections during low blood counts), and the approval was withdrawn in 2024. Tumor shrinkage that does not translate into longer life is exactly the trap careful reading exists to catch. Your low TROP2 result further weakens the case. A trial-only route technically exists, but spending a treatment slot here competes

with better-fitted doors.

Option 14 — Datopotamab deruxtecan

This option was considered but not recommended. This TROP2 drug carries the same payload family as your current drug, which means it re-runs the exact problems already limiting you: the severe nausea, and the lung-inflammation risk that the unresolved chest-scan question makes hard to price. It has no published results in bladder cancer, your TROP2 is low, and back-to-back use of sibling payloads invites the tumor to resist both. The TROP2 door kept open on this page is Option 8, whose payload works differently.

The longer tail: options looked at and set aside

Beyond the fourteen options above, thirty-nine more feature-targeting possibilities were examined and set aside. None was ignored; each carries a recorded reason, and a few are worth knowing about because circumstances could revive them.

Some are blocked by entry rules rather than by science. The most notable is a drug aimed at Nectin-4 and TROP2 at once, blocked solely because its trial excludes anyone with possible drug-induced lung disease, which the unresolved chest-scan discrepancy triggers; a formal lung-specialist review that cleared your scans would convert it into a live candidate, which is one more reason to resolve that discrepancy. Another carries the best HER2 response number in bladder cancer (tumors shrinking in about half of patients) but is built on the payload class that caused your nerve damage, and its US trials exclude prior HER2 treatment besides. A cell-therapy route using tumor-harvested immune cells needs a surgically removable tumor, which you currently do not have, and several others have closed their doors to bladder cancer or have no site a US patient can reach.

Others were folded into options already listed: an alternative attack on the same TP53 mutation as Option 3 that requires manufacturing your own cells and a longer runway, and two backup Nectin-4 drugs kept in reserve if Option 4's screening fails.

A third group rests on targets that only a research-grade test has seen so far: the mesothelin, MUC1, PRAME, and stromal-FAP leads. Each has trials in principle, but each waits on a validated stain or scan that has not confirmed the target, and the record already contains a cautionary tale here: you were immune-type-matched for a MAGE-A4 trial until expression testing came back negative. Expression first, screening second. The related sequencing-dependent ideas (drugs aimed at the ARID1A mutation's repair weaknesses) are hypothesis-grade, and the same rank-one records retrieval doubles as the cheap test of whether any of them apply. Finally, about a dozen concepts fit your tumor on paper but have no open program anywhere a patient can currently reach them.

Questions to ask your oncologist

1. Can we submit the records requests this week for the repeat HER2 result from this month and the full sequencing reports (FGFR3 status, the unnamed actionable alteration, and the HER2 gene copy number), and can the HER2 report state the tumor content of the sample it was read on?
2. What is the concrete plan and timeline for the T-DXd stop-or-continue decision, given the severe nausea after dose 2 and the tumor-DNA blood draws ahead? And if the new HER2 result comes back negative or low, which of the options on this page would we pivot toward?
3. Can we get a pulmonology review to formally resolve the disagreement between last year's chest scan and

June's? It affects the safety pricing of my current drug and would unlock at least one blocked trial.

4. Can we retrieve the June Nectin-4 imaging result before spending any trial-screening visit on a Nectin-4 option, and does any measurable tumor remain on scans after the June radiation, since most of these trials require one?
5. Can we check my existing 2023 sequencing report for the TP53 R175H mutation in tumor tissue, since the trial aimed at it currently rests on one very low-level blood result?
6. Before I continue the vaccine series, can we confirm with the cell-therapy trials we care about that ongoing vaccine dosing will not count against their prior-therapy exclusions?
7. If we pursue the reduced-dose ipilimumab route, what written stopping rules and pancreas monitoring would we put in place, and how do we weigh doing that off-label against saving the slot for a trial?
8. Several trials in this space require prior platinum chemotherapy, which I never had. Does that change how we sequence things from here?

Sources

The recommendations above draw on these published studies and trial registrations.

Publications (PubMed IDs): 27958275, 31100038, 31996390, 32506298, 32568634, 33649166, 33929895, 34385340, 35358455, 35397434, 35749374, 36307410, 36400106, 36868252, 37165196, 37306706, 37385153, 37501016, 37793085, 37870536, 37988648, 38446675, 39086310, 39440991, 39934055, 40081946, 40169866, 40205198, 40346292, 40635151, 40849430, 40862536, 40931013, 41101697, 41173324, 41542775, 41566776, 41686845, 42167230, 42223134

Trial registrations (ClinicalTrials.gov): NCT01174121, NCT02693535, NCT02892123, NCT03228667, NCT03390504, NCT03547973, NCT03686124, NCT03740256, NCT03866382, NCT04262466, NCT04482309, NCT04657068, NCT04879329, NCT04939610, NCT05489211, NCT05568680, NCT05614739, NCT05875168, NCT05877599, NCT05941507, NCT06051695, NCT06241456, NCT06265727, NCT06465069, NCT06518564, NCT06695845, NCT06756035, NCT06778863, NCT06885697, NCT07020117, NCT07156136, NCT07193511, NCT07287995, NCT07419295, NCT07492628, NCT07524348, NCT07545122, NCT07612189, NCT07670312, NCT07686367